Mob. No. 9897529503

Roll No. : .....

P Navneet

MID-SEMESTER EXAMINATION 2025

Name of the Program: **M.Tech** Semester: 3rd

Name of the Course: **Soft Computing** Course Code: **MCS-301**

Time: 90 Minutes **Maximum Marks: 50**

Note:

i. Answer all the questions by choosing any one of the sub-questions.
ii. Each question carries 10 marks

Q1	(10 Marks)	CO1
(a)	Compare and contrast biological neurons and artificial neurons. List the main components of the biological neuron.	
	OR	
(b)	Differentiate between supervised and unsupervised learning. What is the necessity of activation function?	
Q2	(10 Marks)	CO1
(a)	What is the impact of weight in an artificial neural network? Indicate the difference between excitatory and inhibitory weighted interconnections.	
	OR	
(b)	Why is the McCulloch-Pitts neuron widely used in logic functions? Justify-XOR function is non-linearly separable by a single decision boundary line.	
Q3	(10 Marks)	CO1
(a)	How can the equation of a straight line be formed using linear separability? Compare feed-forward and feedback network.	
	OR	
(b)	Using Hebb rule, find the weights required to perform following classifications. The vectors (1-11-1) and (111-1) belong to class (target value +1); vectors (-1-111) and (11-1-1) do not belong to class (target value -1). Also using each of training x vectors as input, test the response of net.	
Q4	(10 Marks)	CO2
(a)	How does learning take place in supervised learning? From a mathematical point of view, what is the process of learning in supervised learning?	
	OR	
(b)	With a neat flowchart, explain the training process of perceptron network.	
Q5	(10 Marks)	CO2
(a)	List the limitations of perceptron. What is the significance of error signal in perceptron network?	
	OR	
(b)	What is meant by incremental learning? Why is gradient descent adopted to minimize error?	